EE3621: DIGITAL SIGNAL PROCESSING | III B.TECH EEE

§ Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

In real-time DSP, input sequences $x[n]$ (speech, audio, radar returns) are effectively infinite in length, while FIR filter impulse responses $h[n]$ have finite length M. Computing the entire convolution at once is impossible due to memory limits and unacceptable latency. The **Overlap-Add (OLA)** method segments the input into non-overlapping blocks of length L and performs block convolution via FFT/IFFT.

Pedagogical Strategy:

1. Formulate input signal decomposition into disjoint blocks: $x[n] = \sum_{m=0}^{\infty} x_m[n - mL]$.
2. Establish block zero-padding requirement: $N \geq L + M - 1$.
3. Compute frequency-domain circular convolution for each block: $y_m[n] = \text{IDFT}_N\{\text{DFT}_N\{x_m\} \cdot \text{DFT}_N\{h\}\}$.
4. Synthesize the total output by adding the overlapping $M - 1$ tails between consecutive blocks.
5. Derive the optimal block length L_{opt} that minimizes total computational cost per output sample.

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Partition** long streaming sequences into blocks suitable for Overlap-Add filtering.
2. **Execute** complete numerical block filtering using the Overlap-Add method.
3. **Determine** the required FFT size N to prevent aliasing within blocks.
4. **Calculate** overall throughput and latency for real-time DSP implementations.

2. MATHEMATICAL FOUNDATIONS

§ 2.1 Overlap-Add Block Decomposition

Let $x[n]$ be an infinite input sequence and $h[n]$ an FIR filter of length M. Segment $x[n]$ into non-overlapping blocks of length L:

$$x_m[n] = \begin{cases} x[n + mL], & 0 \leq n \leq L - 1 \\ 0, & \text{otherwise} \end{cases}$$

Total input: $x[n] = \sum_{m=0}^{\infty} x_m[n - mL]$.

§ 2.2 Linear Convolution of Sub-Blocks

By Linearity of LTI systems:

$$y[n] = x[n] * h[n] = \left(\sum_{m=0}^{\infty} x_m[n - mL] \right) * h[n] = \sum_{m=0}^{\infty} (x_m[n - mL] * h[n]) = \sum_{m=0}^{\infty} y_m[n - mL]$$

Where each block convolution $y_m[n] = x_m[n] * h[n]$ has length:

$$N = L + M - 1$$

§ 2.3 FFT Implementation & Overlap-Addition

1. Choose FFT size $N = 2^k \geq L + M - 1$.
2. Zero-pad $x_m[n]$ with $N - L$ zeros to length N.
3. Zero-pad $h[n]$ with $N - M$ zeros to length N.
4. Precompute $H[k] = \mathrm{DFT}_N\{h[n]\}$.
5. For each block m:
   - $X_m[k] = \mathrm{DFT}_N\{x_m[n]\}$
   - $Y_m[k] = X_m[k] \cdot H[k]$
   - $y_m[n] = \mathrm{IDFT}_N\{Y_m[k]\}$
6. Reconstruct output $y[n]$: Overlap the last $M - 1$ points of block m with the first $M - 1$ points of block $m + 1$ and add.

3. WORKED NUMERICAL EXAMPLES

§ Example 13.1: Complete Numerical Overlap-Add Filtering

Problem: Filter input $x[n] = \{\underset{\uparrow}{1}, 2, -1, 2, 3, -2, 0, 1, 2, 1\}$ using FIR filter $h[n] = \{\underset{\uparrow}{1}, 2, 1\}$ ($M = 3$) using Overlap-Add with block length $L = 4$.

Solution:

- $L = 4,\ M = 3 \implies N = L + M - 1 = 4 + 3 - 1 = 6$. (Pad to $N = 6$).
- Partition input into blocks of $L = 4$:
  - $x_0[n] = \{1, 2, -1, 2, 0, 0\}$ (padded with 2 zeros)
  - $x_1[n] = \{3, -2, 0, 1, 0, 0\}$
  - $x_2[n] = \{2, 1, 0, 0, 0, 0\}$
- Filter padded: $h[n] = \{1, 2, 1, 0, 0, 0\}$.
- **Block Convolutions ($y_m[n] = x_m * h$):**
  - $y_0[n] = \{1, 2, -1, 2\} * \{1, 2, 1\} = \{\underset{\uparrow}{1}, 4, 4, 2, 3, 2\}$
  - $y_1[n] = \{3, -2, 0, 1\} * \{1, 2, 1\} = \{\underset{\uparrow}{3}, 4, -1, -1, 2, 1\}$
  - $y_2[n] = \{2, 1, 0, 0\} * \{1, 2, 1\} = \{\underset{\uparrow}{2}, 5, 4, 1, 0, 0\}$
- **Overlap-Addition ($L = 4$ shift):**

$$\begin{array}{r|cccccccccccc} y_0: & 1 & 4 & 4 & 2 & \mathbf{3} & \mathbf{2} & & & & & & \\ y_1: & & & & & \mathbf{3} & \mathbf{4} & -1 & -1 & \mathbf{2} & \mathbf{1} & & \\ y_2: & & & & & & & & & \mathbf{2} & \mathbf{5} & 4 & 1 \\ \hline y[n]: & 1 & 4 & 4 & 2 & 6 & 6 & -1 & -1 & 4 & 6 & 4 & 1 \end{array}$$

  Result:

$$y[n] = \{\underset{\uparrow}{1}, 4, 4, 2, 6, 6, -1, -1, 4, 6, 4, 1\}$$

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

§ Question 1 (15 Marks)

(a) Describe the Overlap-Add method for linear filtering of long data sequences. Explain why zero-padding is necessary. *(7 Marks)* **(b)** Use the Overlap-Add method to filter $x[n] = \{\underset{\uparrow}{1}, 2, 0, -1, 3, 1, 2, -1\}$ with $h[n] = \{\underset{\uparrow}{1}, 1, 1\}$ using block length $L = 4$. *(8 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Mathematical formulation of block decomposition and overlap additions *(4 Marks)*
  - Explanation of $N \geq L + M - 1$ to prevent circular aliasing *(3 Marks)*
- **Part (b):**
  - Parameter formulation: $L = 4, M = 3 \implies N \geq L + M - 1 = 6$.
  - Block setup:
    - $x_0[n] = \{1, 2, 0, -1, 0, 0\}$ (padded to $N = 6$)
    - $x_1[n] = \{3, 1, 2, -1, 0, 0\}$ (padded to $N = 6$) *(2 Marks)*
  - Block convolutions ($y_m[n] = x_m * h$):
    - $y_0[n] = \{1, 2, 0, -1\} * \{1, 1, 1\} = \{1, 3, 3, 1, -1, -1\}$
    - $y_1[n] = \{3, 1, 2, -1\} * \{1, 1, 1\} = \{3, 4, 6, 2, 1, -1\}$ *(4 Marks)*
  - Overlap-Addition assembly ($L = 4$ shift):

$$\begin{array}{r|cccccccccc} y_0: & 1 & 3 & 3 & 1 & \mathbf{-1} & \mathbf{-1} & & & & \\ y_1: & & & & & \mathbf{3} & \mathbf{4} & 6 & 2 & 1 & -1 \\ \hline y[n]: & 1 & 3 & 3 & 1 & 2 & 3 & 6 & 2 & 1 & -1 \end{array}$$

    Result:

$$y[n] = \{\underset{\uparrow}{1}, 3, 3, 1, 2, 3, 6, 2, 1, -1\}$$

    (2 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

x = np.array([1, 2, 0, -1, 3, 1, 2, -1])
h = np.array([1, 1, 1])
y = np.convolve(x, h)
print("Direct Linear Convolution:", y)
```